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DNA methylation landscape of *Nautilus Pompilius*

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Abstract

Background DNA methylation is a critical epigenetic modification that functions in diverse biological processes. High-throughput sequencing has revealed substantial variations in DNA methylation patterns across species. While extensively studied in vertebrates, knowledge of invertebrate DNA methylation remains limited.

Results We examined the DNA methylation landscape of the ancient-looking marine mollusk, *Nautilus pompilius*, a "living fossil". We identified key DNA methyltransferase genes (*DNMT1* and *DNMT3*) by homologous searches against the *N. pompilius* genome and employed whole-genome bisulfite sequencing (WGBS) to characterize methylation patterns. Despite generally similar gene body methylation (gbM) and genome-wide methylation levels to other invertebrates, the *N. pompilius* exhibits a characteristic low methylation bias specifically in the promoter and first exon regions.

Conclusions *N. pompilius* has a typical gene body methylation pattern. Both promoter and gene body methylation may work together to regulate gene expression in *N. pompilius*. Our research provides valuable information to the ever-expanding methylation map of invertebrates.

Keywords DNA methylation, *Nautilus Pompilius*, Whole-genome bisulfite sequencing, Phylogenetic comparison

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Background

DNA methylation is a critical epigenetic modification involved in diverse biological processes, including gene expression regulation [1, 2], histone modification [3, 4], chromatin structure [5–7], miRNA expression [8, 9], and alternative splicing [10]. High-throughput sequencing has revealed substantial variations in DNA methylation patterns across species [11]. DNA Methylation occurs predominantly at the sequence CpG in animals. Vertebrates typically exhibit genome-wide methylation in CpG sites with notable exceptions in CpG islands, often located within regulatory elements [12, 13]. Methylation in regulatory regions, such as promoters and enhancers, is frequently associated with gene silencing [14, 15]. In contrast, invertebrates display a more varied methylation landscape. Some species, such as *Tribolium castaneum* and *Drosophila melanogaster*, lack DNA methylation

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machinery [11, 16]. Others, including *Apis mellifera* and *Ciona intestinalis*, exhibit lower levels of genome-wide methylation primarily concentrated in CpG sites within gene bodies, known as gene body methylation (gbM) [11, 17]. While the precise role of gbM remains unclear, evidence suggests its association with active gene transcription [11, 17, 18]. Both promoter and gene body methylation contribute to the complex regulation of gene expression [19, 20]. In metazoans, the methylation of cytosines is catalyzed by a family of DNA methyltransferases classified into two groups, namely, DNMT1 and DNMT3 [21]. DNMT1 is often referred to as maintenance methyltransferase since it is thought to be the primary enzyme in charge of copying methylation patterns after DNA replication. DNMT3 are also known as "de novo" methyltransferases, since they do not require hemi-methylated DNA to bind, and they show an equivalent affinity for hemi-methylated and non-methylated DNA [22].

The *Nautilus pompilius* retains numerous archaic features, including an external shell and a simple pinhole eye [23–25]. Thus, research on *N. pompilius* provides important insights into the biology and evolution of cephalopods [26, 27]. Previous studies have revealed that *N. pompilius* possesses the smallest cephalopod genome with exceptionally slow evolutionary rates in both coding and non-coding regions [23, 28]. However, the DNA methylation within cephalopods remains largely unexplored. While limited evidence suggests a functional role for DNA methylation in closely related octopus *Octopus vulgaris* [29] and squid *Doryteuthis pealeii* [30], a comprehensive understanding of DNA methylation in *N. pompilius* is lacking. Investigating DNA methylation in *N. pompilius* not only fills a critical knowledge gap but also provides a unique opportunity to explore the evolution and innovation of DNA methylation within cephalopods.

In this study, we conducted whole-genome bisulfite sequencing (WGBS) on *N. pompilius* to characterize its DNA methylation patterns and investigate the relationship between DNA methylation and gene expression. Our data revealed typical gene body methylation in *N. pompilius*, with correlations observed between both promoter and gene body methylation and gene expression. These results may offer novel insights into DNA methylation in invertebrates.

Methods

Sample collection

In October 2016, a juvenile *Nautilus pompilius* with a shell diameter of 12 cm was collected in the vicinity of the Nansha Islands within the South China Sea. The individual utilized in the present study was the same specimen as that described in our previous publication [23]. Following collection, the specimen was transported in a temperature-controlled, light-impermeable tank, where the temperature was maintained within a range of 16–19 °C to ensure optimal preservation during transit. Subsequently, this individual was generously donated by the Chinese Ocean Conservation Association for the specific purposes of this research investigation. The whole genome of this specimen has been already sequenced (Genbank no. GWHBECW00000000) in our previous study [23]. DNA was extracted from muscle, mantle, eye, tentacle, funnel, gill, beak, crop stomach and liver tissues using a DNeasy Blood & Tissue Kit (QIAGEN). DNA quantity and quality were assessed via agarose gel electrophoresis and Qubit 2.0 fluorometry (Thermo Fisher Scientific).

Research ethics

Prior to the commencement of any anatomical dissection, complete and irreversible euthanasia of the *N. pompilius* individual was rigorously confirmed. Upon arrival, the donated nautilus was observed to be in a moribund state, characterized by a markedly diminished responsiveness to external stimuli. To definitively ascertain the absence of neurological function and life, the specimen was immersed in a 2% ethanol-seawater solution [31], and the complete abolition of reflex activity was meticulously monitored. Specifically, the absence of any motor or sensory response to acupuncture and the complete lack of reaction to tactile stimulation were carefully and repeatedly assessed. Tissue sample collection was initiated only after the unequivocal confirmation of the complete cessation of all vital signs. All experimental protocols were reviewed and approved by the Research Ethics Committee for Animal Experiments at the South China Sea Institute of Oceanology, Chinese Academy of Sciences (SCSIO2025009).

Library construction and bisulfite sequencing

Whole-genome bisulfite sequencing libraries were constructed using the Acegen Bisulfite-Seq Library Prep Kit (Acegen, Cat. No. AG0311) according to the manufacturer's protocol. Briefly, 1 μg of genomic DNA spiked with 1 ng of unmethylated lambda DNA was fragmented by sonication to a mean size of approximately 200–500 bp, then end-repaired, 5'-phosphorylated, 3'-dA-tailed and ligated to 5-methylcytosine-modified adapters. After bisulfite treatment, the DNA was amplified via 10 cycles of PCR using Illumina 8-bp dual index primers. The constructed WGBS libraries were then analyzed by an Agilent 2100 Bioanalyzer and finally sequenced on Illumina platforms using a 150 × 2 paired-end sequencing protocol.

Quality control and mapping

WGBS data for *C. intestinalis* and *N. vectensis* were obtained from accession number GSE19824 [11], while WGBS data for *C. gigas* were obtained from accession number GSE40302 [32]. Initially, the raw reads were processed by adapter removal and filtering via Trimmomatic (v0.39) [33] utilizing the following parameters: ILLUMINACLIP: adapter.fa:2:30:10:2:keepBothReads LEADING:3 TRAILING:3 MINLEN:36 HEADCROP:9. The resulting reads that successfully passed all filtering criteria were regarded as clean reads. To ensure the quality of the clean data, FastQC (v0.11.9) [34] was used for basic quality control. Clean reads were aligned to the reference genomes of *C. intestinalis* (GCF_000224145.3), *N. vectensis* (GCF_932526225.1), *C. gigas* (GCF_902806645.1), and *N. pompilius* (GWHBECW00000000) [23] using Bismark (v 0.23.1) [35]. The bismark_genome_preparation command was used to prepare the reference genome. The bismark_methylation_extractor command was used to extract methylation information from the alignment results. Finally, the bismark2bedGraph command was used to report the methylation results for each cytosine site with the parameter -zero_base.

CpG methylation level

We used the weighted methylation level in covered CpGs to quantify the CpG methylation level in a particular region. The weighted methylation level (ML) is defined as:

$$\mathrm{ML} = \sum\nolimits_{i=1}^{n} Ci / \sum\nolimits_{i=1}^{n} (Ci + Ti)$$

where C = reads supporting methylated cytosine, T = reads supporting unmethylated cytosine, i = position of CpGs, and n = total number of CpG positions. Only CpGs with coverage > 4 were counted. The above process is implemented in Python.

Transcriptome data analysis

The RNA-seq data were downloaded from the SRA database with accession nos. SRR11485678–SRR11485687 and then trimmed based on quality scores using Btrim v.0.2.0. Clean reads were aligned to the *N. pompilius* genome with TopHat v.2.1.1 [36]. Gene abundance in different tissues was calculated using Cufflinks v.2.1.1 [37].

Identification of DNA methyltransferases

The identification of DNA methyltransferases (DNMTs) followed the methodology described by Engelhardt et al. (2022) [38]. Specifically, the predicted proteins of *N. pompilius* were downloaded from the GSA database (accession no. GWHBECW00000000) [23]. The protein domain models for DNA_methylase (PF00145, code in Pfam), ADD_DNMT3 (PF17980), CH (PF00307), PWWP (PF00855), BAH (PF01426), DMAP_binding (PF06464), DNMT1-RFD (PF12047) and zf-CXXC (PF02008) were downloaded from the "Pfam protein families database" [39, 40]. Initially, only the DNA_methylase model was used to identify DNA methyltransferase candidates in the set of proteins predicted using hmmsearch from HMMER software (v 3.3) [41]. Proteins with a predicted DNA methylation domain and a full sequence e-value < 0.001 were further considered candidates. A DNMT1 candidate was required not to have a PWWP or ADD_DNMT3 domain; and at the same time, have a DNMT1_RFD, zf-CXXC and BAH domain. When a protein fulfills these two criteria, it was considered a full Dnmt1. A DNMT3 candidate, on the other hand, was required not to have a DNMT1_RFD, zf-CXXC or BAH domain, while have both a PWWP domain and an ADD_DNMT3 domain. Proteins have these two features were considered a full Dnmt3.

Phylogenetic analysis

DNA methyltransferase sequences from the studied species which were related to *N. pompilius* were obtained from NCBI [42] and UniProt [43] (details in Table S1). First, the sequences were aligned by ClustalW [44], after which the best maximum likelihood (ML) models were computed. For this study, we opted for the "LG + G + I + F" model. The phylogenetic tree was constructed by default parameters. Moreover, we used the bootstrap method to estimate the reliability of the tree, with the number of bootstrap replicates set to 1000.

Identification of tissue-specific genes

We used the Tau index to measure tissue-specific genes [45]. The index tau is defined as:

$$Tau = \sum\nolimits_{i=1}^{N} (1 - Xi)/(N - 1)$$
$$Xi = FPKMi/\max(FPKMi)$$

where N is the number of tissues and FPKMi is the FPKM in those tissues.

Tau ranges from 0 (expressed ubiquitously) to 1 (tissue-specific). We selected genes with Tau ≥ 0.8 and average FPKM ≥ 0.3 as tissue-specific genes.

Statistical analysis

Data visualization and statistical analyses were performed using GraphPad Prism 8.0. Statistical significance was determined at the 0.05 level (*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$). Correlation analyses were conducted in R (version 3.6.3).

Results

The *N. pompilius* genome contains genes encoding DNA methyltransferases (DNMTs)

To investigate the existence of the DNMT proteins in *N. pompilius*, homologous searches using DNA methylase domain models against the *N. pompilius* predicted proteins [23] were conducted, and one ortholog each for DNMT1 and DNMT3 was identified (Fig. 1A and B). In detail, we found that in *N. pompilius*, both DNMT1 and DNMT3 have a methylation catalytic site (PF00145, code in Pfam). In addition to the common sites, DNMT1 contained a DNMT1-RFD domain (PF12047), a zf-CXXC domain (PF02008) and two BAH domains (PF01426); while DNMT3 contained an ADD_DNMT3 domain (PF17980) and a PWWP domain (PF00855) (Fig. 1A and B). To further check whether the proteins we identified are indeed DNMTs, samples from multiple representative animals were selected, and phylogenetic trees were constructed based on their DNA methyltransferases' sequences (Fig. 1C and D). The phylogenetic tree we obtained were consistent with traditional taxonomy, particularly regarding the relationships among the three cephalopods: *N. pompilius*, *Octopus vulgaris*, and *Octopus bimaculoides*. This result indicated that the DNMT1 and DNMT3 we identified in *N. pompilius* are truly DNA methyltransferases.

Subsequently, we assessed the expression levels of these two *DNMT* genes in various tissues of *N. pompilius* using our previously published RNA-seq data from our previous study [23]. The data showed that the FPKM of

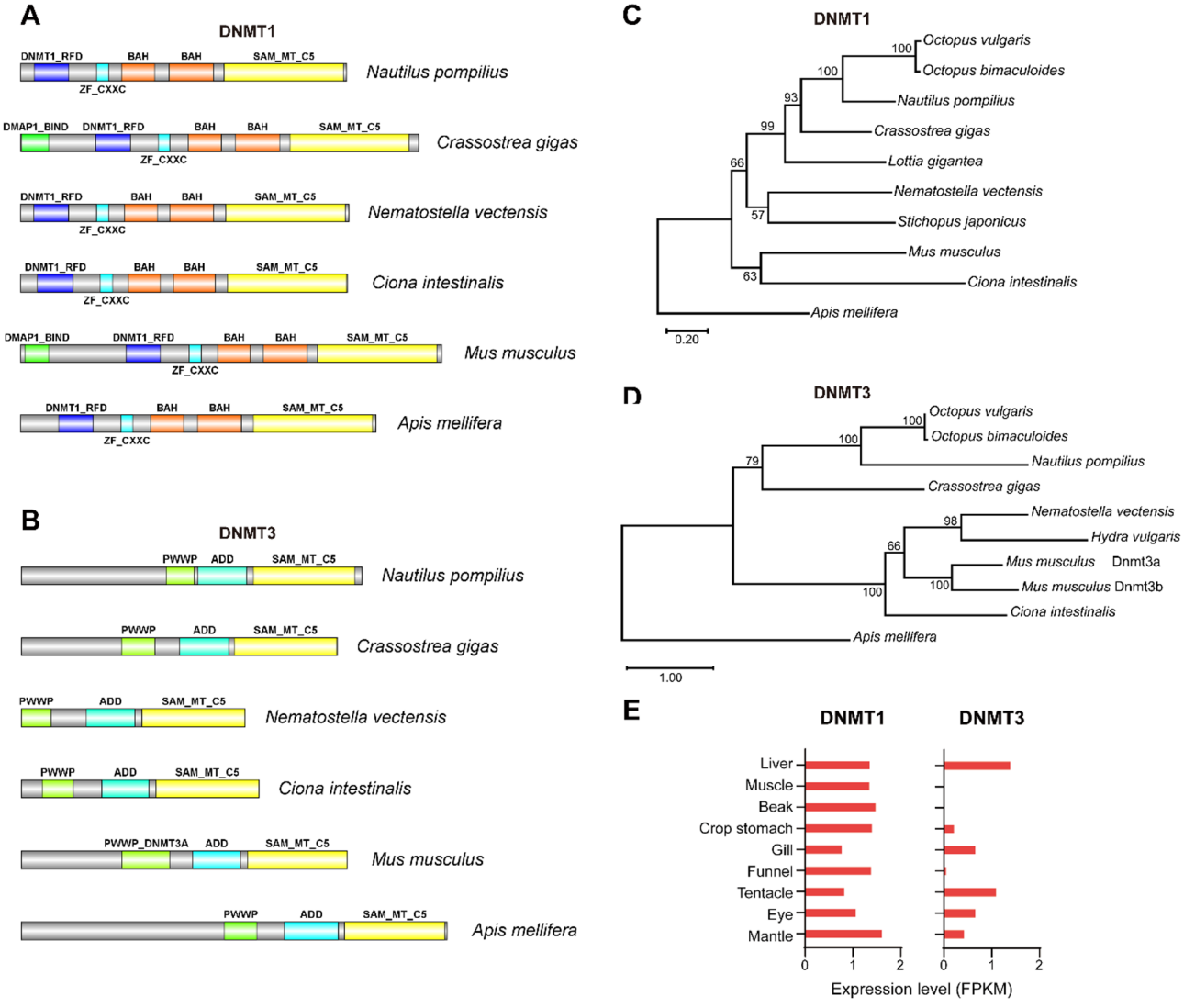

Fig. 1 Characterization of DNA methyltransferases (DNMTs) in *Nautilus pompilius*. (**A**, **B**) Conserved domain organization of DNMT1 and DNMT3 proteins in *N. pompilius* compared to *Crassostrea gigas*, *Nematostella vectensis*, *Ciona intestinalis*, *Mus musculus*, and *Apis mellifera*. (**C**, **D**) Phylogenetic analysis of DNMT1 and DNMT3 protein sequences across species. (**E**) Tissue-specific expression patterns of *DNMT* genes in *N. pompilius* based on RNA-seq data from our previous study [23]

DNMT1 gene was greater than 1 in the liver (1.41), muscle (1.40), beak (1.43), crop stomach (1.46), funnel (1.44), eye (1.12), and mantle (1.67), and was close to 1 in the gill (0.83) and tentacle (0.88). In contrast, the FPKM of *DNMT3* gene was greater than 1 only in the liver (1.45) and tentacle (1.15), while it was very low in the muscle (0.05), beak (0.08), and funnel (0.11) (Fig. 1E). Our results indicated that *DNMT1* is expressed broadly across different tissues with similar expression levels, while *DNMT3* showed more tissue-specific expression pattern.

Genome-wide methylation level in *N. pompilius*

Our whole-genome bisulfite sequencing of the 9 samples (raw sequence data have been deposited in the Genome Sequence Archive [46, 47], CRA016930) yielded 100.5 million reads on average, and 90.4 million of these reads (90%) were clean data. Approximately 55.3% of the clean reads were uniquely mapped to the reference genome with an average depth of approximately 21×. The conversion rates of all 9 samples were greater than 99.6%. The relevant details of all the samples' data are listed in Table S2. To assess genome-wide methylation levels, we compared the results to those of other marine metazoans (*Crassostrea gigas*, *Nematostella vectensis*, and *Ciona intestinalis*) and representative vertebrates (*Homo sapiens* and *Mus musculus*) [11, 17, 32, 48]. We found that overall genome methylation in *N. pompilius* (2.44%) was intermediate, exceeding that of oyster (1.71%) and sea anemone (1.79%), but lower than that of sea squirt (4.62%), human (3.93%), and mouse (4.15%) (Table S3). Particularly, CpG methylation levels followed a similar trend, with *N. pompilius* (21.2%) exhibiting values comparable to that of sea squirt (23.37%), but surpassing that of oyster (10.48%) and sea anemone (9.38%). In addition, the CHH (0.39%) and CHG (0.42%) methylation levels were negligible in *N. pompilius* and other marine metazoans examined, contrasting with plants where these modifications play crucial roles [17, 19, 49]. Therefore, our further analysis focused exclusively on CpG methylation in *N. pompilius*.

Gene body methylation patterns in *N. pompilius*

To determine whether *N. pompilius* has gene body methylation, we analyzed its gene methylation landscape. Our results revealed prominent gene body methylation in *N. pompilius* (Fig. 2A and B, Figure S1-S8), consistent with typical invertebrate models. In addition, gene body methylation levels exceeded those of the genome-wide average (21.21%) (Fig. 2B and C). Conversely, promoters (defined as 500 bp upstream of the transcription start site) and the first exons were generally unmethylated (Fig. 2B, D and E), with a strong positive correlation between their methylation levels (Fig. 2F). For instance, the gene *EVMG014464* displayed typical gene body methylation (blue bar in Fig. 2A) with an unmethylated promoter region (grey bar) (Fig. 2A). To investigate the relationship between DNA methylation and gene expression, we categorized genes based on expression levels (FPKM) and analyzed methylation patterns across gene elements (Fig. 2B). The genes were categorized into four groups based on their expression levels: the first group had FPKM values greater than 100, the second group had FPKM values between 100 and 10, the third group had FPKM values between 10 and 0.3, and the fourth group had FPKM values less than 0.3. Genes in the fourth group also meant unexpressed genes. Results showed that moderately expressed genes exhibited the highest gene body methylation, while unexpressed genes showed moderate levels of gene body methylation (Fig. 2C), aligning with previous invertebrate studies [11, 17, 32]. For the expressed genes, promoter and the first exon methylation negatively correlated with gene expression (Fig. 2D and E). However, when considering all genes, the correlation between promoter methylation and gene expression was weak and negative (Spearman's rho = -0.05, $P < 10^{-11}$), while gene body methylation showed a general positive correlation with gene expression (Spearman's rho = 0.32, $P < 10^{-15}$). To further explore this relationship, correlation analyses were conducted for each gene in nine different tissues. The results showed that promoter methylation was negatively correlated with the expression of 2725 genes (absolute value of Spearman's correlation coefficient ≥ 0.3, 211 of which were significantly negatively correlated ($P < 0.05$)), and 2592 genes were positively correlated (258 of which were significantly positively correlated). Similarly, gene body methylation levels showed negative correlations with the expression of 2,878 genes, 275 of which showed significant negative correlations. In contrast, they were positively correlated with the expression of 2,989 genes, 328 of which showed significant positive correlations. Thus, there is no consistent trend in the effect of methylation on the expression of specific genes across different tissues. These findings suggest a complex interplay between DNA methylation and gene regulation, likely with both activating and inhibitory roles for both promoter and gene body methylation.

Evolution of methylation in *N. pompilius* and other marine invertebrates

To elucidate the evolution of methylation patterns, we compared gene body methylation profiles among *N. pompilius* and other marine invertebrates-*Crassostrea gigas*, *Nematostella vectensis*, and *Ciona intestinalis* [11, 32]. Our analysis revealed that the methylation levels of gene in *C. gigas* and *N. vectensis* were similar to those in *N. pompilius*. Furthermore, the *N. pompilius* and *C. gigas* exhibited similar methylation patterns, particularly within gene bodies, although overall methylation

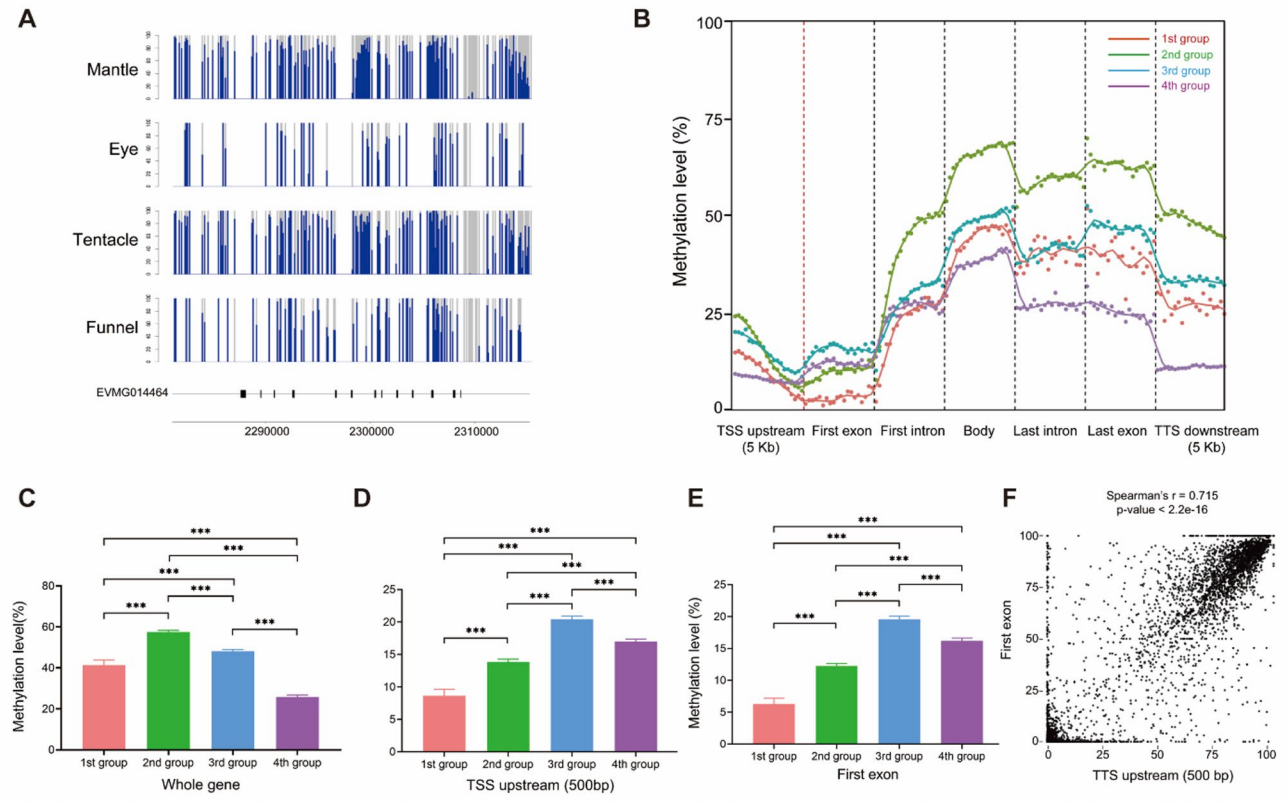

Fig. 2 Genomic distribution of mCG methylation in *N. pompilius*. (**A**) mCG methylation profile of the *EVMG014464* gene located on Contig02233 (2285343–2308677, minus strand). Each bin represents 100 bp. Gray bars indicate CpG sites with sequencing coverage ≥ 4, while blue bars represent average methylation levels. Data shown for mantle, eye, tentacle, and funnel tissues. (**B**) Distribution of mCG methylation across gene elements in mantle tissue. Genes were categorized into four expression groups based on FPKM values: >100, 100 – 10, 10 – 0.3, and < 0.3. The canonical gene structure is divided into seven features. The length of each feature was normalized and divided into equal numbers of bins. Dots represent mean methylation per bin, lines connect locally weighted scatterplot smoothing curves. The red vertical line indicates the transcription start site (TSS). (**C**, **D**, **E**) Average methylation levels for each gene element in different groups. Error bars represent standard error of the mean (SEM). Asterisks indicate significant differences (***$P < 0.001$). (**F**) Correlation between promoter and first exon methylation levels for individual genes. The horizontal and vertical coordinates represent the average methylation level

levels were higher in *N. pompilius* (Fig. 3A, B and C). In contrast, *C. intestinalis* displayed a distinct pattern, resembling vertebrates with lower gene body methylation (Fig. 3D), consistent with previous findings [11, 50]. Notably, all these examined species demonstrated a conserved preference for unmethylated promoter regions. Thus, the similar methylation patterns across invertebrates highlight their strong evolutionary conservation.

Relationship between tissue-specific gene expression and DNA methylation in *N. pompilius*

Given the potential role of DNA methylation in tissue-specific gene regulation [51, 52], we investigated methylation patterns of tissue-specific genes using above mentioned RNA-seq data. Re-analysis of the RNA-seq from our previous study [23] identified a total of 1424 tissue-specific genes across nine tissues of *N. pompilius* (Fig. 4A). To illustrate, mantle-specific genes exhibited significantly lower methylation levels compared to other genes across most gene elements (Fig. 4B). Therefore, to simplify the presentation of the results, the various elements of the genes were combined for statistical analysis (Fig. 4D). This hypomethylation pattern extended to mantle-specific genes examined in other eight tissues (Fig. 4C, E and F), and the other tissue-specific genes also had the same hypomethylation pattern. Importantly, we examined the expression of these tissue-specific genes and found that they contained genes with low, medium or high expression, which meant that this hypomethylation was independent of gene expression levels and suggesting a general characteristic of tissue-specific genes in *N. pompilius*.

Discussion

Due to the unique ecological characteristics and limited population size of *N. pompilius*, in-depth studies have long been constrained. In this study, we show that in *N. pompilius* there are *DNMT1* and *DNMT3* genes which

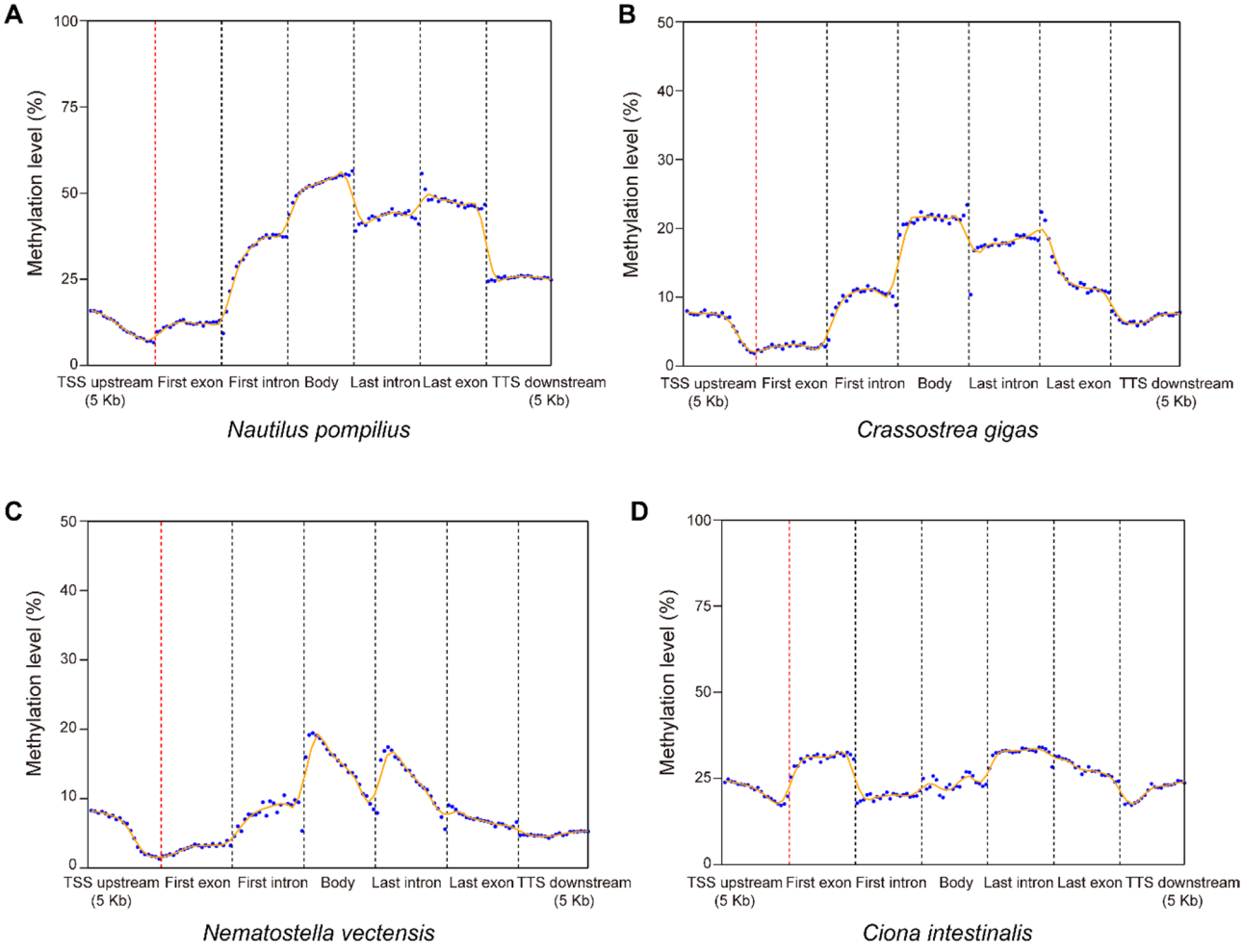

Fig. 3 Comparative DNA methylation patterns in several marine invertebrates. mCG methylation profiles across gene elements in (**A**) *N. pompilius* (mantle), (**B**) *C. gigas* (mantle), (**C**) *N. vectensis* (whole adult), and (**D**) *C. intestinalis* (muscle). Gene structures are divided into seven features. Dots represent mean methylation per bin, lines connect locally weighted scatterplot smoothing curves. The red vertical line indicates the transcription start site (TSS)

were identified by homology searches using DNA methyltransferase domain models in the predicted *N. pompilius* proteome, and that the whole genome bisulfite sequencing of 9 tissues of *N. pompilius* following comparative analysis reveal that the methylation patterns of the *N. pompilius* are consistent with those of other invertebrates.

Both DNMT1 and DNMT3 proteins we identified in the *N. pompilius* exhibited conserved domains characteristic of their respective families. Phylogenetic analysis, including sequences from multiple species also elucidated their evolutionary relationships. These findings demonstrate that DNMT proteins are evolutionarily conserved in various species, including *N. pompilius*. RNA-seq data revealed broad tissue expression of DNMT1 with relatively consistent levels, while DNMT3 exhibited a more tissue-specific expression pattern in *N. pompilius*. Notably, the CpG methylation level in all examined invertebrates, including *N. pompilius*, was substantially lower than that in vertebrates, consistent with previous studies [11]. While invertebrate methylation levels vary widely (2–80%) [53, 54], *N. pompilius* aligns with the majority of invertebrates having under 40% CpG methylation levels. Furthermore, the methylation patterns of genes in *C. gigas* and *N. pompilius* are similar, suggesting that DNA methylation is highly conserved across cephalopods and bivalves.

In *N. pompilius*, we observed predominant gene body methylation with relatively hypomethylated promoters, a pattern commonly found in invertebrates. Consistent with previous findings in other organisms, we found that promoter methylation exhibited a weak negative correlation with gene expression in *N. pompilius* [14, 15, 20, 55]. However, the relationship between promoter methylation and gene expression for specific genes is intricate, as evidenced by both positive and negative correlations observed in various species and cellular contexts [50, 52, 56, 57]. In our research, we also identified specific genes

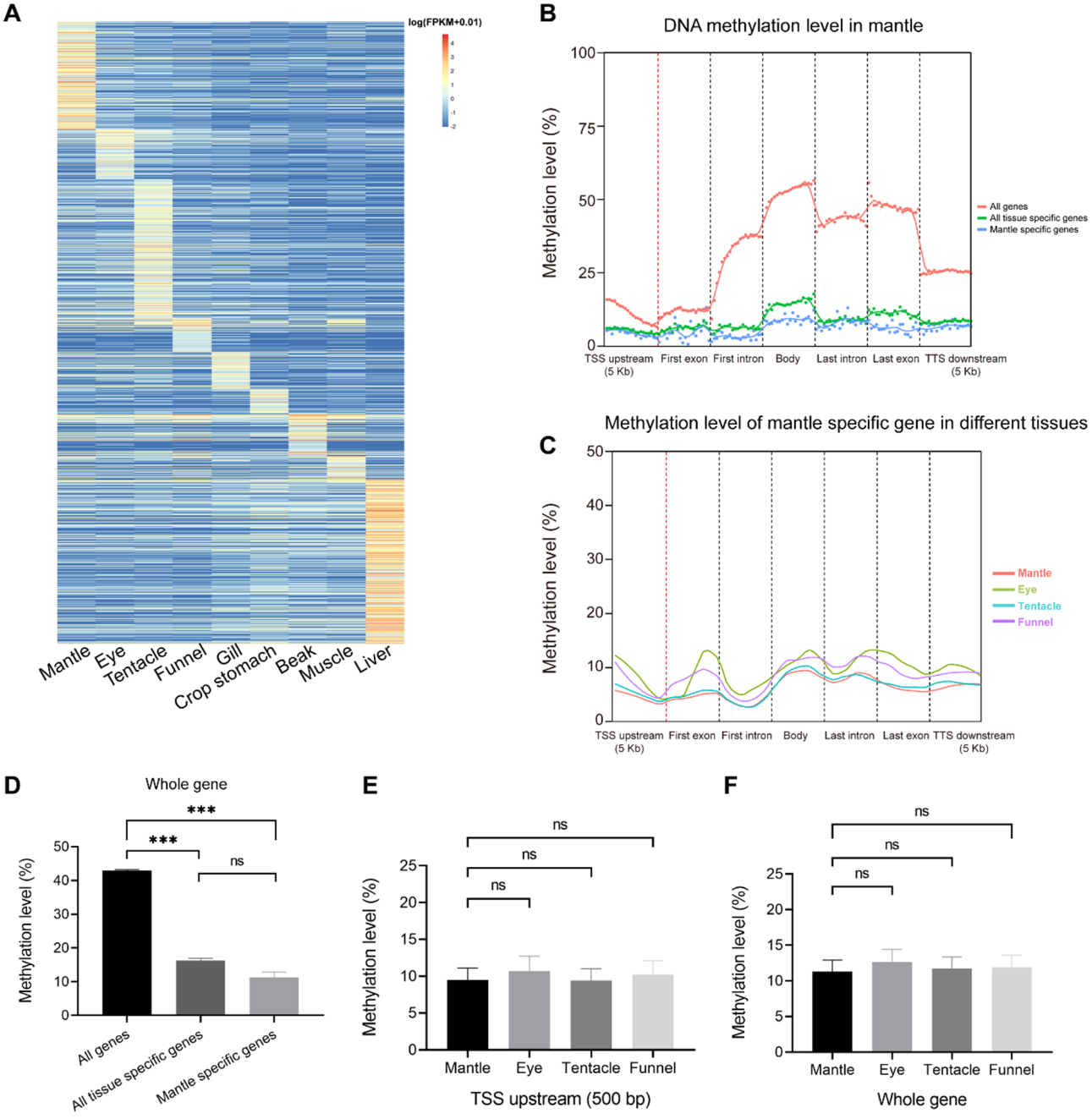

Fig. 4 DNA methylation patterns of tissue-specific genes in *N. pompilius*. (**A**) Heatmap of 1424 tissue-specific genes (FPKM ≥ 0.3) across nine tissues of *N. pompilius*. (**B**) Comparison of mCG methylation levels in all genes, tissue-specific genes, and mantle-specific genes within gene elements in mantle of *N. pompilius*. All genes ($n = 17709$), tissue-specific genes ($n = 1424$), and mantle genes ($n = 245$) are shown (red, green, blue lines respectively). (**C**) mCGs of mantle-specific genes for various genetic elements in four tissues of *N. pompilius*. (**D**) Gene body methylation levels in the mantle in the three groups indicated. (**E**) Promoter methylation levels of mantle-specific genes in four tissues indicated. (**F**) Gene body methylation levels of mantle-specific genes in four tissues indicated. Error bars represent standard error of the mean (SEM). Asterisks indicate significant differences (***$P < 0.001$)

that exhibit either positive or negative correlations with promoter methylation and their expression levels in various tissues, a phenomenon that has been observed in human cells [52]. Our results thus contribute to this complex picture by demonstrating both positive and negative correlations between promoter methylation and gene expression in *N. pompilius*.

While first exon methylation may link to gene silencing in vertebrates [58], its role in invertebrates remains unclear. Our findings suggest a potential link between first exon hypomethylation and gene expression in *N.*

pompilius. However, the strong correlation with promoter methylation complicates the interpretation of these results, which suggests that the methylation of the first exon may be a continuation of promoter methylation.

Gene body methylation, frequently associated with active transcription in invertebrates [11, 17, 18], exhibited a positive correlation with gene expression in *N. pompilius*. Interestingly, moderately expressed genes displayed the highest levels of gene body methylation. While these findings suggest a general association between gene body methylation and gene expression, the complex interplay between methylation and gene regulation is evident in the tissue variations observed. Surprisingly, our study identified a novel pattern of low gene body methylation in tissue-specific genes in *N. pompilius*. These findings indicate a potential involvement of gene body methylation in regulating tissue-specific gene expression, although additional mechanisms likely contribute too. This observation warrants further investigation to elucidate its underlying mechanisms and potential functional significance.

The strength of our study lies in being the first to explore the epigenetic landscape of the *N. pompilius*, a rare and precious species. This addresses the knowledge gap in epigenetic research on cephalopods. However, several limitations should be also taken into consideration. First, WGBS cannot differentiate between 5-methylcytosine (5mC) and 5-hydroxymethylcytosine (5hmC). The potential presence of 5hmC in *N. pompilius* may lead to overestimation of 5mC methylation levels. Second, our study was based on a single specimen, which may introduce bias due to individual variation. The study would benefit from including more samples. However, due to the scarcity, this was not feasible at this moment, and future studies would consider this in collaboration with other researchers. Finally, our findings rely on sequencing data and lack validation through molecular biological experiments. With advancing technologies and the collection of additional specimens in the future, we may gain deeper insights into *N. pompilius*, broadening the research horizons in cephalopod.

Abbreviations

DNMT	DNA methyltransferase
WGBS	Whole-genome bisulfite sequencing
gbM	gene body methylation
TSS	Transcription start site
FPKM	Fragments per kilobase of transcript per million mapped reads
SEM	Standard error of the mean

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12864-025-11907-0.

Supplementary Material 1

Author contributions

X.S., Y.Z. and H.M. conceived and designed the experiment; Y.W., H.M. and F.M. conducted the experiment; Y.W. and H.M. carried out analyses, visualized and interpreted results; Z.Z., X.D. and Z.Y. provided resources and supervision. All authors wrote, reviewed and gave the final approval for publication of the manuscript.

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Data availability

The raw sequence data reported in this paper have been deposited in the Genome Sequence Archive [55] of the National Genomics Data Center [56], China National Center for Bioinformation/Beijing Institute of Genomics, Chinese Academy of Sciences (GSA: CRA016930), which are publicly accessible at https://ngdc.cncb.ac.cn/gsa.

Declarations

Ethics approval and consent to participate

The study was approved by the Ethics Committee of Chinese Academy of Science with approval no. of SCSIO2025009.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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